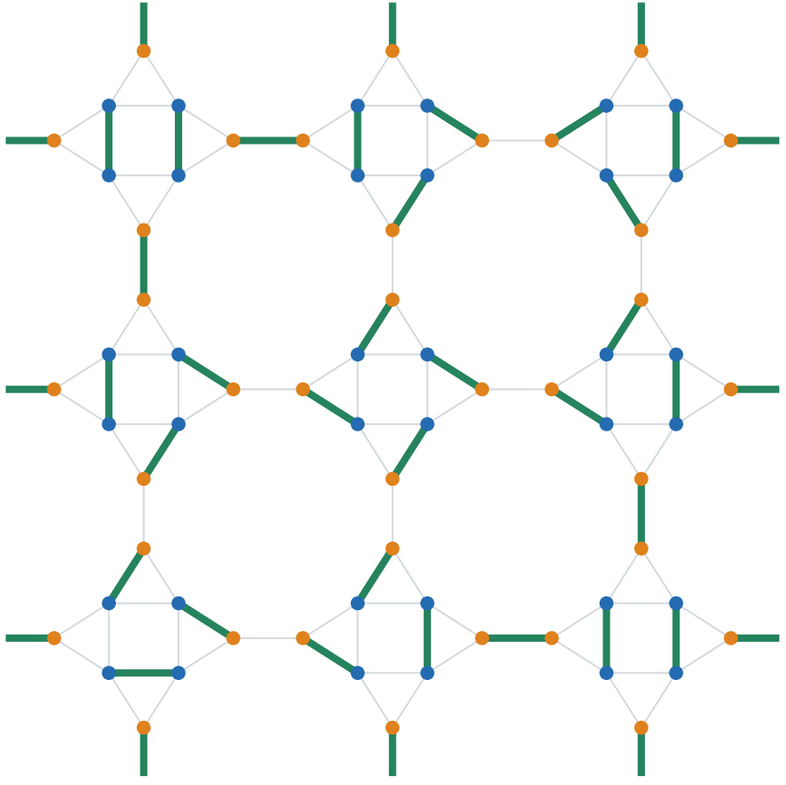
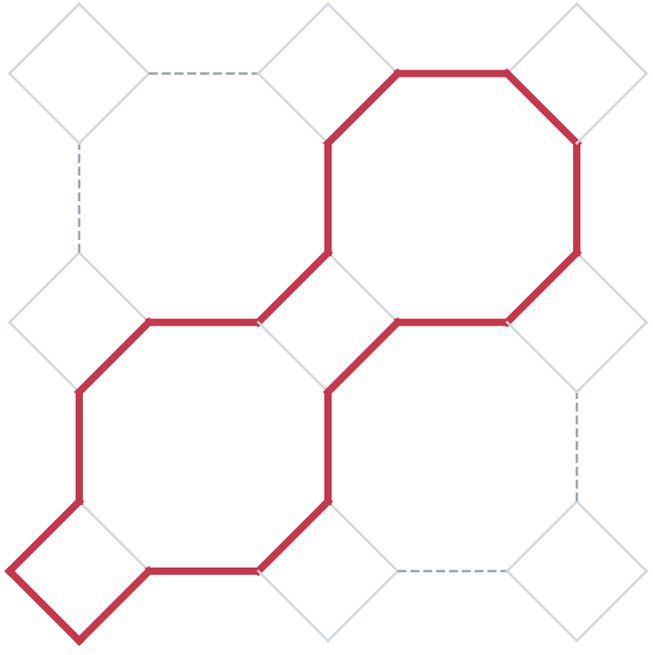
Your mapping: covering → square–octagon loops → contract dummy links

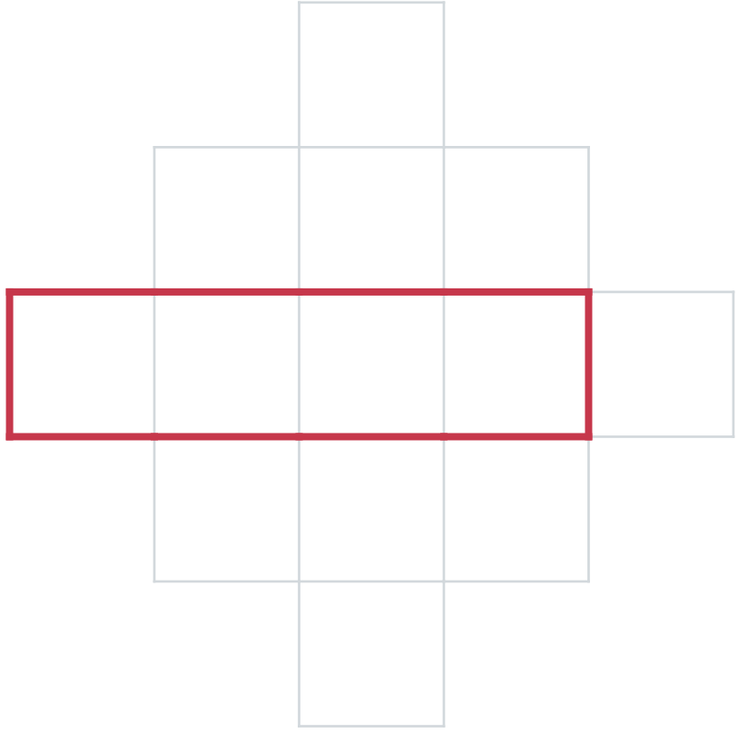
1. One allowed singlet covering



2. Loops on the square–octagon graph



3. Shrink dummy links → square-lattice loops



The last step is one-to-one, with the inherited loop routing. It does not collapse the small squares.